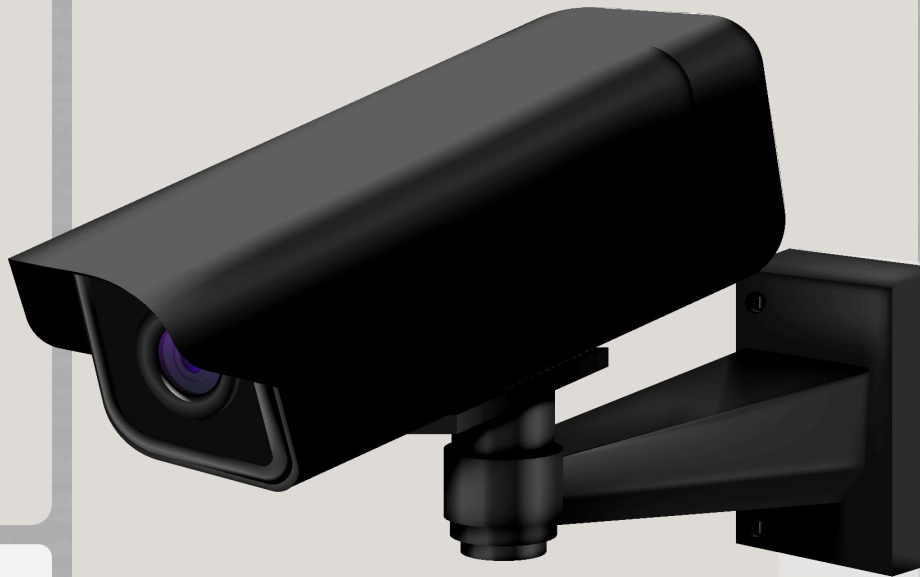


Is Big Brother always watching?

(Surveillance systems of
America)

By Robin Hurtado and Amy Tran



Introduction:



Government agencies use a variety of technologies to monitor and collect information on the public. These range from aerial devices, to body cameras, to police cars which automatically scan and capture license plate numbers that come into view. They capture location, date, times, sound and pictures, sending all this information into a central server. Government officials are constantly promoting these different technologies for public safety, but what are the possible consequences and threats to our privacy?

Our questions:

- Some questions we have are what types of technologies are being used?
- Which agencies are using them?
- What type of technologies is each type of law enforcement likely to use?



Our Datasets:

1. *Atlas of Surveillance:*

- From the Electronic Frontier Foundation
- Represents different U.S. law-enforcement agencies, the surveillance technologies they use
- The unit of observation is a single technology used by an agency and each column measures the details of the agency and Technology.

2. *Automated License Plate Reader Dataset:*

- From the Electronic Frontier Foundation
- Unit of observation is ALPR data from agencies.
- 2016-2017
- Cars scanned by ALPR and how many of the vehicles were deemed "hot."
- Includes links to reports.
- How data was cleaned:
 - States were formatted, capitalized and abbreviated
 - Null or empty values were given the place holder name "Not provided."

3. *Atlas of Surveillance: Southwestern Border Communities*

- From the Electronic Frontier Foundation and University of Nevada, Reno
- Unit of observation is agencies from border states, such as California or Texas, and their technology usage
- How data was cleaned:
 - Blank counties filled with Census area
 - Counties lists in a list were separated

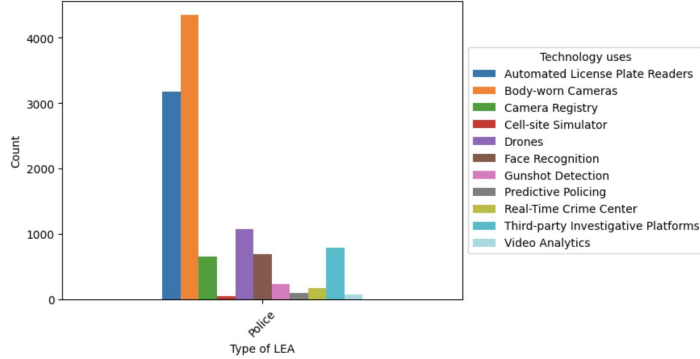
4. *Scholars Under Surveillances*

- 2021, University of Nevada, Reno, Student report
- Unit of observation is agencies from colleges, and their technology usage
- Problems:
 - How do we clean blank/ null columns



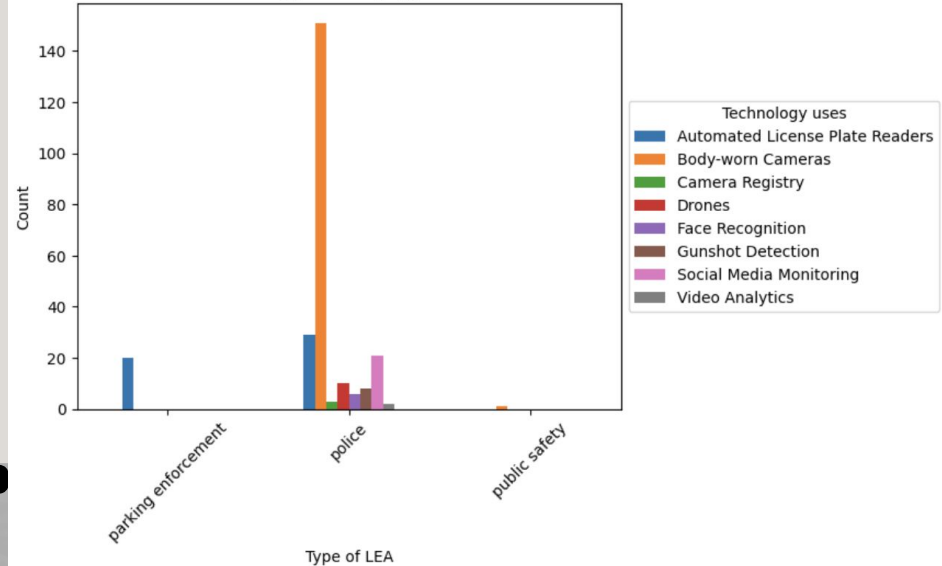
Visuals

Technology count for only the police for All states

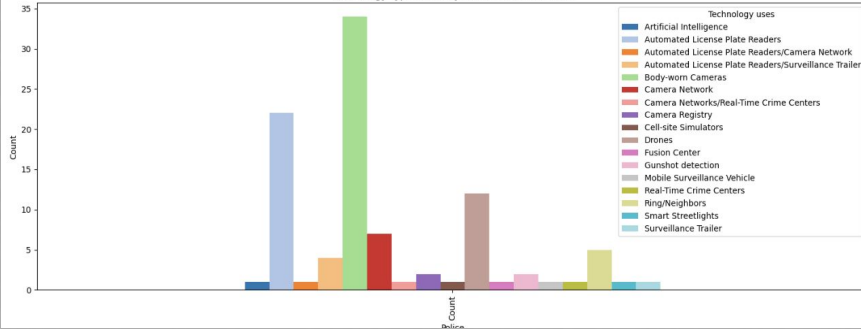


Technology usage from the police departments across the U.S

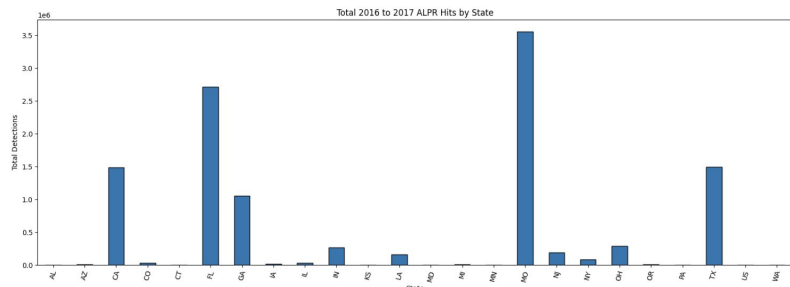
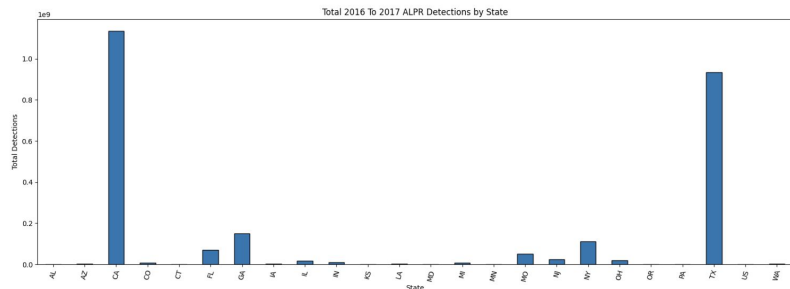
Technology count by Type of LEA



Technology Types Used by Police in Border States



Visuals



ALPR Data U.S states (Detections and hits)

- The amount of License plates that were scanned from 2016 to 2017
- And those who were flagged as being a stolen vehicle, registration expired, etc.

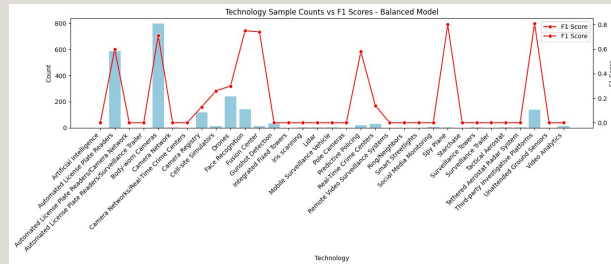
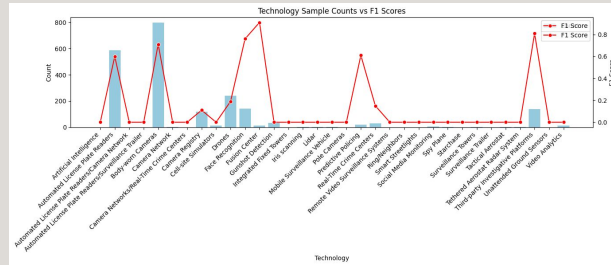
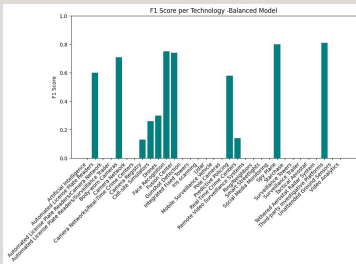
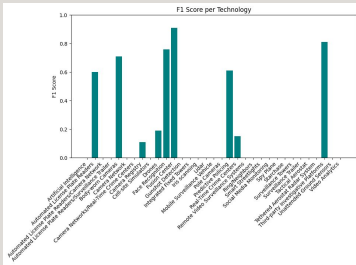
Visuals

Surveillance at SDSU

	Ref	Agency	City	County	State	TypeofLEA	TypeofJuris	Technology	Vendor	Summary
0	EDU016	California State University San Marcos Police ...	San Marcos	San Diego County	CA	police	University	Body-worn Cameras	Axon	California State University San Marcos Police ...
1	EDU112	Palomar College Police Department	San Marcos	San Diego County	CA	police	University	Automated License Plate Readers	None	The Palomar College Police Department began us...
2	EDU124	San Diego State University Police Department	San Diego	San Diego County	CA	police	University	Automated License Plate Readers	None	The San Diego State University Police Departme...
3	EDU125	San Diego State University Police Department	San Diego	San Diego County	CA	police	University	Body-worn Cameras	None	The San Diego State University Police Departme...
4	EDU163	University of California San Diego Police Depa...	La Jolla	San Diego County	CA	police	University	Body-worn Cameras	None	As of 2020, all University of California, San ...
5	EDU164	University of California San Diego Police Depa...	San Diego	San Diego County	CA	police	University	Gunshot Detection	ShotSpotter	The University of California San Diego Police ...
6	EDU165	University of California San Diego Transportat...	San Diego	San Diego County	CA	parking enforcement	University	Automated License Plate Readers	Park Assist	The University of California San Diego has use...

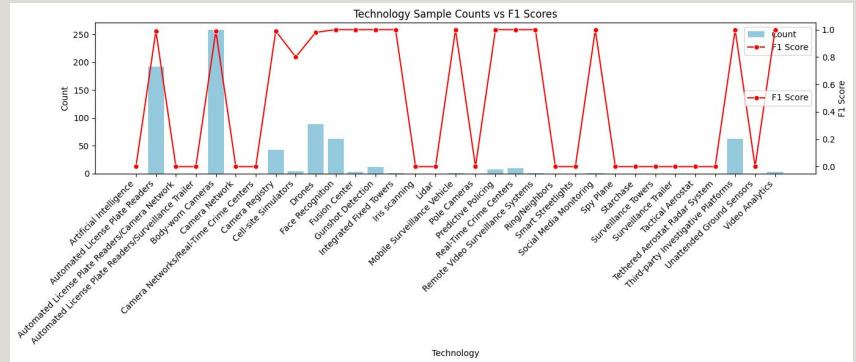
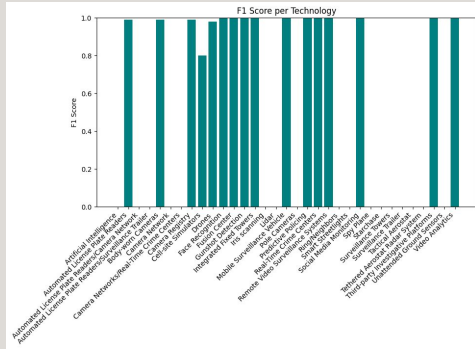
5

Model Selection:



11

Final Model Weighted:





Some Difficulties:

Do we need all of our datasets?:

- Many repeat information
- See what columns to remove or keep
- Some datasets not organized to compliment rest of data

County inconsistency?:

- Some listed individually
- Some listed with multiple counties
- We were not able to split those rows into counties without either duplicating data or guessing at how the data split by the counties listed, so we had to shift our analysis to the city level instead.

Cleaning numeric values?:

- Null values from our ALPR dataset were given string placeholder names
- When cleaned the data could not be summed up
- Had get rid of the commas to proper sum up values
- ALPR data is always changing, differences in the way data is cleaned?



Some more Difficulties:

Typos and Errors:

- Difference in characters used “single quote” vs “apostrophe”
- Spacing, spelling accuracy

When to say when?:

- District vs Office
- Expanding large Jurisdictions by city
- What is a borough?



Decisions we made:

1. Should we include data sharing?

- Our ALPR Dataset has information on data sharing, which we considered analyzing, but many of the links to the data sharing information led to documents that had been converted to images.
- Manually extracting the data would have taken more time than given for the project, and attempting to extract the text using Computer Vision would have required another project and is beyond the scope of this class.

2. Should we combine similar technology types?

- Huge list of technologies
- Should we combine them
 - Ex: Automated License plate readers and Automated License plate readers/Camera Network
- We decided not to combine them since most of them seem to be distinct and different



Decisions we made pt2:

3. Should observe only a few states?

- Our 3 datasets have Arizona, California, and Texas represented so should we focus on only those three?
- We ended up keeping all the data and make our results more general.

4. What loss function and model?

- Our data is categorical
- We will use F1 scores
- Decision Trees will work best
- To weight or not to weight



Thank You!

